Massey University
College of Sciences

161.304: Comments and Solutions for Computer Laboratory 3

Exercise 1: Boys and Girls

1. 3H1 Grid approximation for the probability of a birth being a boy
   We assume a prior distribution for the sought probability First create a grid and specify the prior:

   ```
   grid.resol <- 1000
   p_grid <- seq(from = 0, to = 1, length.out = grid.resol)
   prior <- rep(1,  grid.resol)
   ```

   Now count the boys and the total number of kids

   ```
   boys <- sum(birth1)+sum(birth2)
   kids <- length(birth1)+length(birth2)
   ```

   And then compute the likelihood for (explanation) probability value on the grid and the posterior:

   ```
   lik.boys <- dbinom(boys, kids, prob = p_grid)
   posterior <- lik.boys*prior
   posterior <- posterior/sum(posterior)
   ```

   And we can plot the posterior we obtain (it is a Beta distribution, feel free to check that it is a Beta(1+boys, 1+girls) because of conjugate prior) and find the mode of the posterior:

   ```
   plot(p_grid, posterior, lwd = 3, col = "blue", type = 'l', xlab = "Proportion of Boys",
      ylab = "", main = "Posterior density")
   p_grid[which.max(posterior)] # this is the mode
   ```

2. 3H2 Posterior samples and HPDI

   ```
   samples.prop.boys <- sample(p_grid, prob = posterior, size = 10000, replace = TRUE)
   HPDI(samples.prop.boys, prob = c(0.50, 0.89, 0.97))
   ```

3. 3H3 10000× boy count simulations from 200-birth experiments

   ```
   boys.count <- rbinom(10000, size = 200, prob = samples.prop.boys)
   dens(boys.count, adj = 0.6, show.HPDI = 0.9, xlab = "Number of boys",
      ylab = "", main = "Distribution of simulated boy counts among 200 births")
   abline(v = boys, lty = 2, col = "red", lwd = 2)
   #And you could add an HPDI manually
   ```

   What matters is that based on these simulations, the model seems to be very compatible with the 111 observed count, right in the middle of the distribution. Again, no surprise as we know that the prior won't influence the posterior much, so the posterior is mostly driven by the data/the likelihood. So it is not a discovery, more an internal consistency check.

   Notice that the `adj` option is only here to make the density plot smoother. Change its value to see what happens!

4. 3H4 Same work but with first born kids only

   Not clear if we can reuse the `samples.prop.boys` from 3H2 or if we should redo the exercise working on the `birth1` data only.

   I will re-use `samples.prop.boys` from 3H2. It's quicker :). But also "the model" has to refer to the posterior on the proportion of boys we estimated from the entire population.

```
boys.count.first <- rbinom(10000, size = 100, prob = samples.prop.boys)
dens(boys.count.first, adj = 0.8, , show.HPDI = 0.89,  xlab = "Number of boys"
   , ylab = "",
   main = "Distribution of simulated boy counts among 100 births")
abline(v = sum(birth1), lty = 2, col = "red", lwd = 2)
HPDI(boys.count.first, prob = c(0.5, 0.9))
```

   Slightly less central. But, we still fit in the 50% HPDI so not that different.

   Notice that if we were to re-fit the probabilities on the first born only, the model would probably fit much better. Do it for the sake of practising!

   Some more explanations in 3H5!

5. 3H5 Influence on first born sex on second born sex

   We will build a model that simulates sex of second birth when the first birth is a female assuming it has no influence on it. And compare it to the data in our sample.

```
girls1 <- length(birth1)-sum(birth1)
born2 <- rbinom(10000, size = girls1, prob = samples.prop.boys)
boy.born.after.first.girl <- sum(birth2[which(birth1 == 0)])
dens(born2, adj = 0.8, , show.HPDI = 0.89, xlab = "Number of boys", ylab = "",
   main = "Distribution of simulated boy births when first born is girl")
abline(v = boy.born.after.first.girl, lty = 2, col = "red", lwd = 2)
HPDI(born2, prob = c(0.5, 0.9, 0.99))
```

   Now we are quite far, outside the 99% HPDI: the model strongly under-estimates the number of second born boys when the first born is a girl.

   A good explanation is that the independence of sex at birth assumption is probably violated.